EDUCATION PROJECT'S REPORT

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1. Introduction

Educational opportunity is not spread evenly across U.S. communities. Household income, employment, education levels, and family structure can all be linked to how well students do in school. This project asks how well socioeconomic factors predict average ACT scores at U.S. high schools, and whether knowing a school's charter status adds anything once those factors are accounted for.

I begin with exploratory analysis of the distributions and correlations in the data. I then fit regression models in Python with statsmodels: simple linear and quadratic models that use median household income alone, and a multiple linear regression that combines five socioeconomic predictors with a charter-status indicator. The goal is prediction and description, not estimating causal effects.

2. Data and Preparation

2.1 Data sources

The first file, EdGap_data.xlsx, has 7,986 observations and seven fields on school ACT performance and socioeconomic conditions. EdGap pulls its socioeconomic indicators from the U.S. Census Bureau's American Community Survey and its test results from public state education sources. The second file, ccd_sch_029_1617_w_la_11212017.csv, comes from the National Center for Education Statistics (NCES) Common Core of Data for 2016–2017. It has 102,183 observations and 65 fields and supplies school identifiers and characteristics, including charter status. EdGap warns that testing requirements differ by state, which limits direct comparison of average scores across states.

2.2 Data cleaning and merging

In education_data_preparation.ipynb, I loaded the Excel and CSV files with pandas, checked dimensions, data types, and missing values, and kept the relevant school and socioeconomic fields. I converted the school identifiers to a common format and merged the two datasets with a left join. I then selected high schools, checked for duplicate records and implausible values, set invalid ACT scores and lunch percentages to missing, and dropped records with no valid outcome. Missing numerical predictors were filled in with scikit-learn's IterativeImputer. Finally, I renamed the columns to lowercase snake_case and exported the tidy, school-level table to data/education_clean.csv.

The cleaned file contains 7,227 school records, and all of them are used in the models below. The charter variable has three values: 6,858 schools are labeled "No," 170 "Yes," and 199 "Not applicable." The predictors are the unemployment rate (rate_unemployment), college attainment (percent_college), percent married (percent_married), median household income (median_income), and the share of students receiving free or reduced-price lunch (percent_lunch). The percentage variables are stored as proportions between 0 and 1.

3. Exploratory Data Analysis

In education_eda.ipynb, I used pandas, Seaborn, and Matplotlib to examine the correlation matrix of the numerical variables, pair plots colored by charter status, and boxplots for spotting outliers. I used these to judge the direction and rough shape of relationships, see how strongly the predictors overlap, and look for unusual values.

Every socioeconomic variable was associated with average ACT score, and the relationships looked roughly linear. The strongest was school lunch participation, with Pearson's $r \approx -0.785$: schools where a larger share of students received free or reduced-price lunch tended to have lower average ACT scores. The other four predictors were more modest, with correlations of about 0.43 to 0.46 in absolute value (0.46 for median income and for college attainment, 0.44 for percent married, and −0.43 for unemployment). These are associations in the data, not evidence that any one factor causes differences in scores.

The predictors also overlap with one another. Pairwise correlations among them range from about 0.41 to 0.71 in absolute value, with the strongest being median income and college attainment (0.71) and median income and percent married (0.62). This overlap suggests that not every predictor will add much once the others are in the model, and it means individual regression coefficients should not be read as independent effects. Boxplots showed outliers in the unemployment rate, college attainment, percent married, and median income. None looked obviously incorrect; they appear to reflect skewed distributions, so I kept them. Log transformations of these predictors could be worth trying in later models. The pair plots showed similar relationships for charter and non-charter schools, although the fitted lines might differ.

4. Regression Methods

The response variable was average_act, and I fit every model with ordinary least squares using the statsmodels formula interface. First I fit median income alone as a simple linear regression and as a quadratic regression, then compared the two with an F-test (anova_lm) to see whether the squared term helped. Next I fit a multiple linear regression with five socioeconomic predictors (rate_unemployment, percent_college, percent_married, median_income, and percent_lunch) and charter status entered as a categorical variable with C(charter).

Charter status has three levels, so the model uses indicator variables for "Yes" and "Not applicable," with "No" as the reference category. In education_eda.ipynb, every model was fit to all 7,227 schools, so those fit statistics describe how well each model matches these data. The five-predictor model was also evaluated with a train/test split in the modeling notebook: I split the data 80/20 using train_test_split(test_size=0.2, random_state=42), fit scikit-learn's LinearRegression on the training data, and scored it on the held-out test set. I report R^2, root mean squared error (RMSE), and mean absolute error (MAE). Lower RMSE and MAE mean smaller prediction errors, and R^2 is the share of variation in ACT scores that the model accounts for.

The multiple regression has the form $ACT_i = \beta_0 + \beta_1\ unemployment_i + \beta_2\ college_i + \beta_3\ married_i + \beta_4\ income_i + \beta_5\ lunch_i + \beta_6\ notapplicable_i + \beta_7\ charter_yes_i + \varepsilon_i$, where the last two terms indicate charter = "Not applicable" and charter = "Yes." The research question asks whether charter status improves predictions, so I added it alongside the socioeconomic measures.

5. Results

Model	Evaluation	R^2	MAE	RMSE
Median income only (linear)	In-sample	0.211	1.713	2.228
Median income only (quadratic)	In-sample	0.219	1.697	—
Five socioeconomic predictors + charter	In-sample	0.630	—	—
Five socioeconomic predictors	Test set	0.615	1.161	1.554

Table 1. Fit statistics for the regression models. In-sample rows use all 7,227 schools; the test-set row uses the 20% held-out sample. MAE and RMSE are in ACT points; a dash means that statistic was not computed. The two evaluation methods are not directly comparable.

Median income alone explained about 21% of the variation in ACT scores ($R^2 = 0.211$), with an RMSE of 2.228 and an MAE of 1.713 ACT points. The slope of 4.7×10^{-5} per dollar works out to roughly 0.47 ACT points per $10,000 of median income. Adding a squared term raised R^2 slightly to 0.219 and lowered the MAE to 1.697. The F-test favored the quadratic model ($F = 73.2$, $p \approx 1.4 \times 10^{-17}$), and its negative squared coefficient indicates that the benefit of higher income levels off. The practical gain is small, though: MAE fell by only about 0.02 ACT points.

The multiple regression fit much better, with $R^2 = 0.630$ (adjusted $R^2 = 0.630$) across all 7,227 schools. School lunch participation was by far the strongest predictor. Its coefficient of −7.60 ($t = -78.7$) means that a school with a lunch share 10 percentage points higher is predicted to score about 0.76 ACT points lower, holding the other variables fixed. Unemployment rate (−2.31) and college attainment (+1.75) were also significant ($p < 0.001$). Percent married ($p = 0.623$) and median income ($p = 0.572$) were not significant once the other variables were included, which is consistent with the

overlap among predictors seen in the correlation matrix: median income explained a fair amount on its own but added little here.

On the held-out test set, the five socioeconomic predictors reached $R^2 = 0.615$, MAE = 1.161, and RMSE = 1.554 ACT points. Adding charter status did not noticeably change these test-set metrics, so it did not improve prediction accuracy in my comparison. The test R^2 and the in-sample R^2 of 0.630 are similar in size, but they come from different models and evaluation methods, so I do not compare them directly.

For charter status in the full-data model, schools labeled "Yes" scored about 0.19 points lower than schools labeled "No" on average after accounting for the socioeconomic variables, but the difference was not statistically significant (95% CI −0.42 to 0.05, $p = 0.115$). Schools labeled "Not applicable" scored about 0.69 points lower than "No" schools ($p < 0.001$). In this model, then, charter status showed no clear evidence that charter schools differ from non-charter schools in ACT scores. I did not fit a version of the full-data model without charter status, so the test-set comparison above is the clearer evidence on accuracy.

6. Critical Evaluation

6.1 Did the research question address the original problem?

Yes. The research question asks whether socioeconomic conditions can predict average ACT scores, which speaks directly to educational inequality and could be investigated with the available school performance and socioeconomic data.

6.2 Which socioeconomic factor was the strongest predictor?

School lunch participation had the strongest individual correlation with average ACT scores ($r \approx -0.785$), larger in absolute value than the correlation for median household income ($r = 0.460$). One possible explanation is that free or reduced-price lunch participation reflects the economic circumstances of the students attending a school more directly than median household income does. Median household income is measured at the census tract level and may include households with no students at the school, so lunch participation may be a more direct measure of economic disadvantage among the student population. However, the strongest individual correlation does not necessarily mean that a variable has the largest independent contribution in a multiple regression. In the full-data regression, lunch participation did also have the largest t-statistic, while median income was not significant once the other predictors were included.

6.3 How well did the models answer the research question?

The multiple regression model achieved a test R^2 of 0.615, suggesting that the socioeconomic variables together provide meaningful predictive information about average ACT scores. However, the model does not fully explain differences in school performance. The median-income models also showed that a single predictor carries less explanatory information than several socioeconomic factors considered together (R^2 of 0.211 and 0.219, compared with 0.615 on the test set and 0.630 in-sample), although those results were obtained using different evaluation methods.

6.4 What were the limitations of the data?

Several limitations should be considered when interpreting the results:

- Some observations contained missing or invalid values, which required cleaning and imputation.
- Census tract socioeconomic characteristics may not accurately represent the backgrounds of individual students.
- Differences in ACT participation and testing policies across states may affect comparisons.
- Charter schools make up a relatively small share of the dataset (170 of 7,227 schools), and 199 schools have charter status "Not applicable," a category whose meaning should be confirmed in the CCD documentation.
- The imputation was performed before the train/test split, which may introduce information leakage into the reported test results.
- The models in education_eda.ipynb were fit and assessed on the same schools, and the residuals of the multiple regression are not normally distributed (skew of 0.57, kurtosis of 6.0). The large condition number (about 1.3×10^6) reflects correlated predictors and the very different scales of the variables, so individual coefficients should be interpreted with caution.

- The results are based on observational data and cannot establish causal relationships.

These limitations should be considered when interpreting the results and generalizing the findings.

7. Conclusion

This project investigated whether socioeconomic factors can predict average ACT scores among U.S. high schools. The analysis found that socioeconomic factors are associated with school performance. Free or reduced-price lunch participation had the strongest individual correlation with average ACT scores, while median household income also showed a positive relationship.

The multiple regression model achieved a test R^2 of 0.615, suggesting that the selected socioeconomic factors explain a substantial portion of the variation in average ACT scores in the testing dataset. Adding charter school status did not noticeably improve prediction accuracy. Overall, the findings suggest that socioeconomic conditions are useful for predicting average ACT scores, but they do not fully explain differences in academic performance.

Future research could include additional school-level characteristics, such as enrollment, school funding, teacher experience, and access to educational resources.

References

EdGap. (n.d.). About the EdGap map and data. https://www.edgap.org/

National Center for Education Statistics. (n.d.). Common Core of Data (CCD). https://nces.ed.gov/ccd/